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PROGRAMME DE RECHERCHE
MATÉRIAUX ÉMERGENTS

Introduction to workflows

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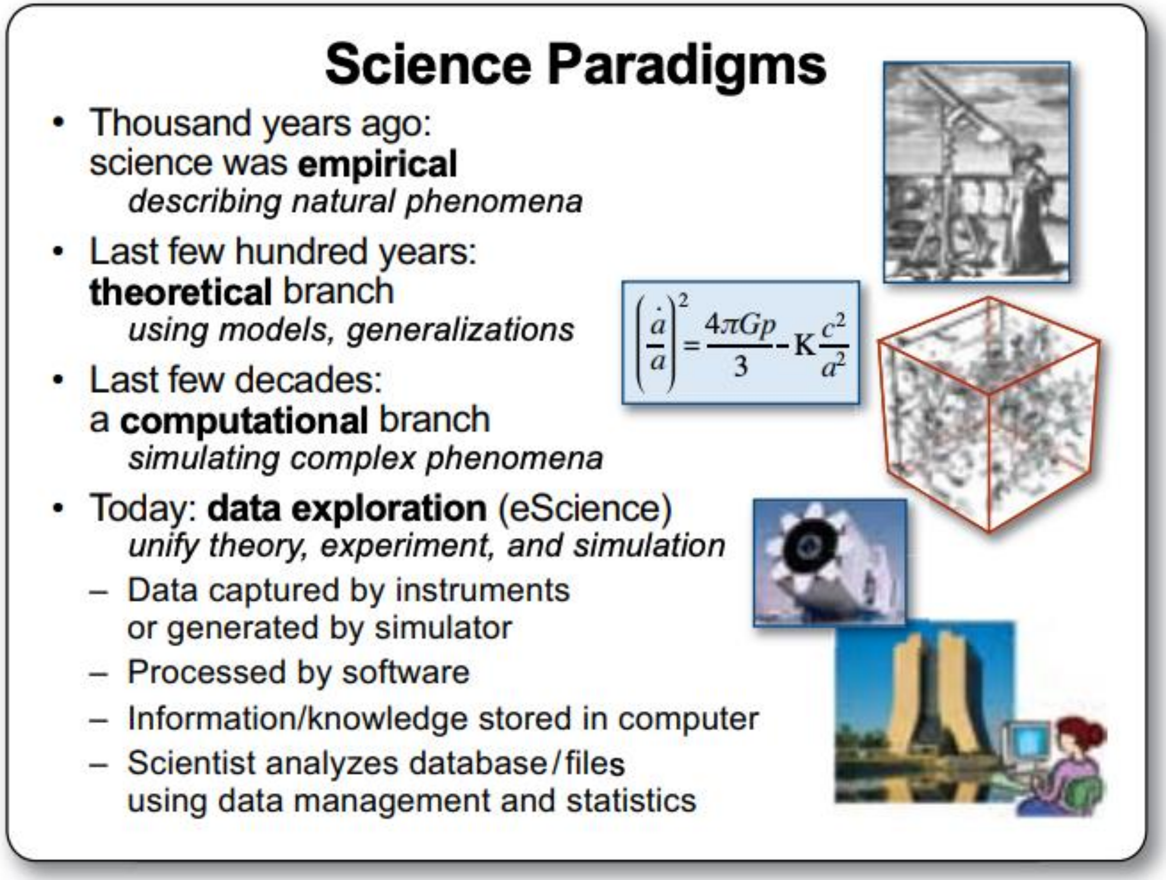
Outline

- Understand workflow fundamentals
- Survey established workflow management systems
- Hands-on **AiiDA workflow** on google colab.

Introduction

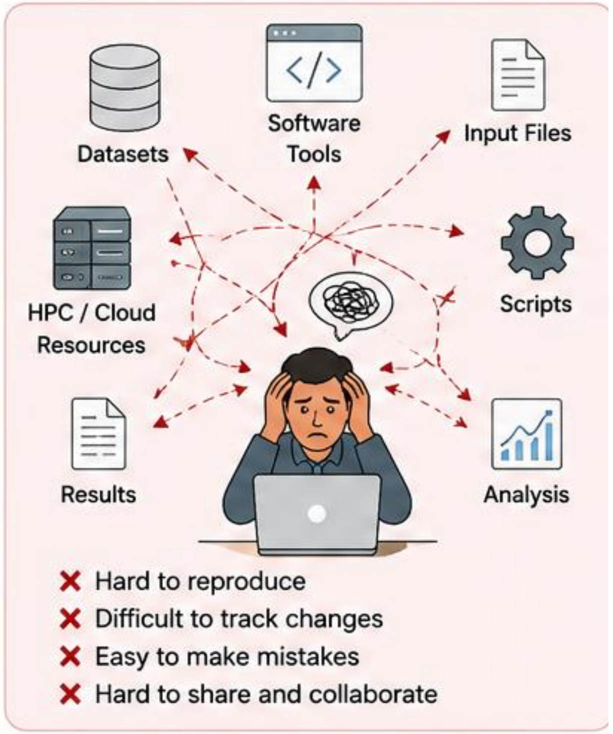


Fourth paradigm is using the computational power to generate, curate, analyze, archive massive amounts of scientific data

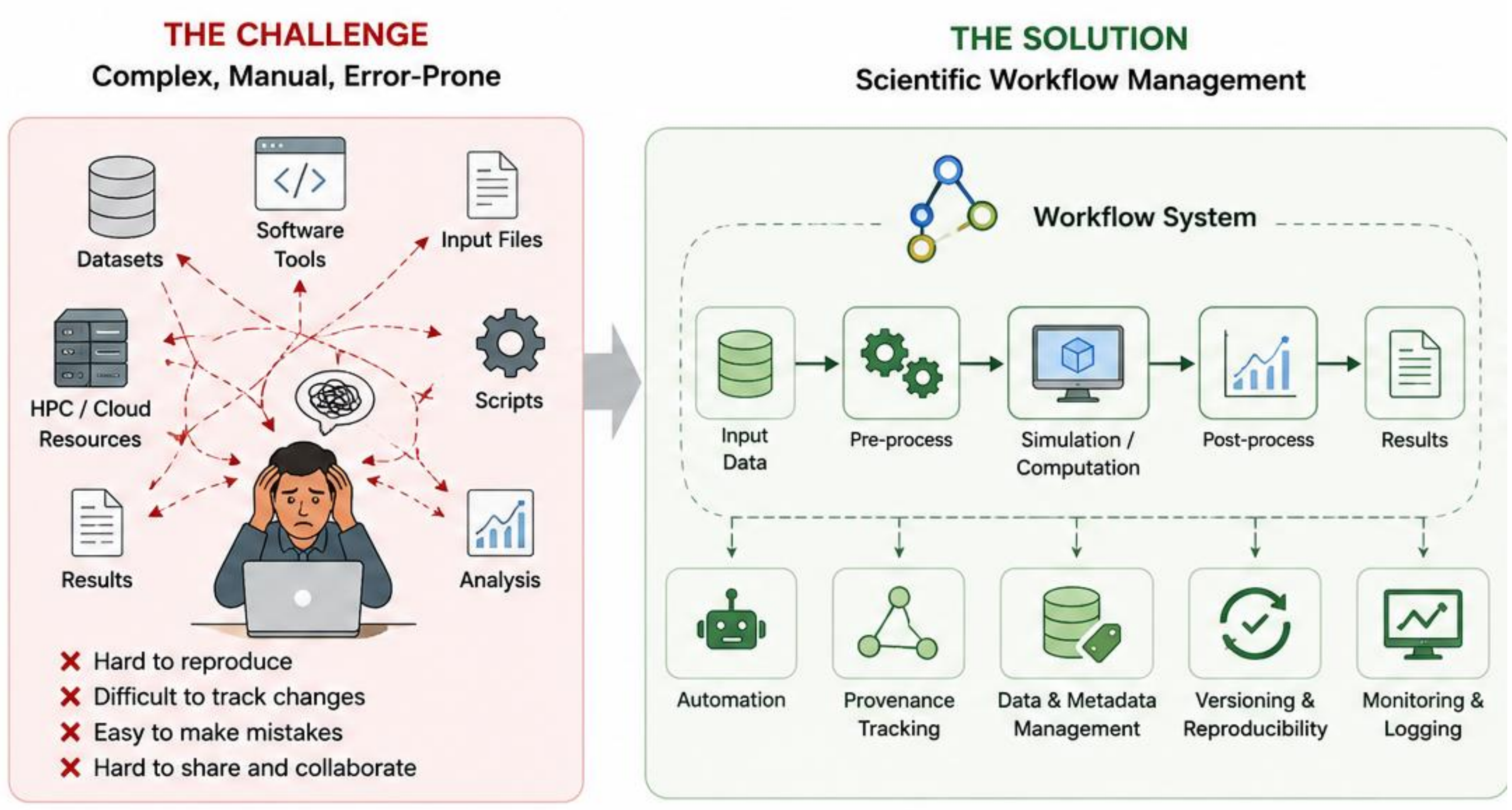
Introduction

- Modern scientific research involves
  - many simulations,
  - datasets,
  - software tools
  - processing steps

- Manually managing everything becomes difficult.

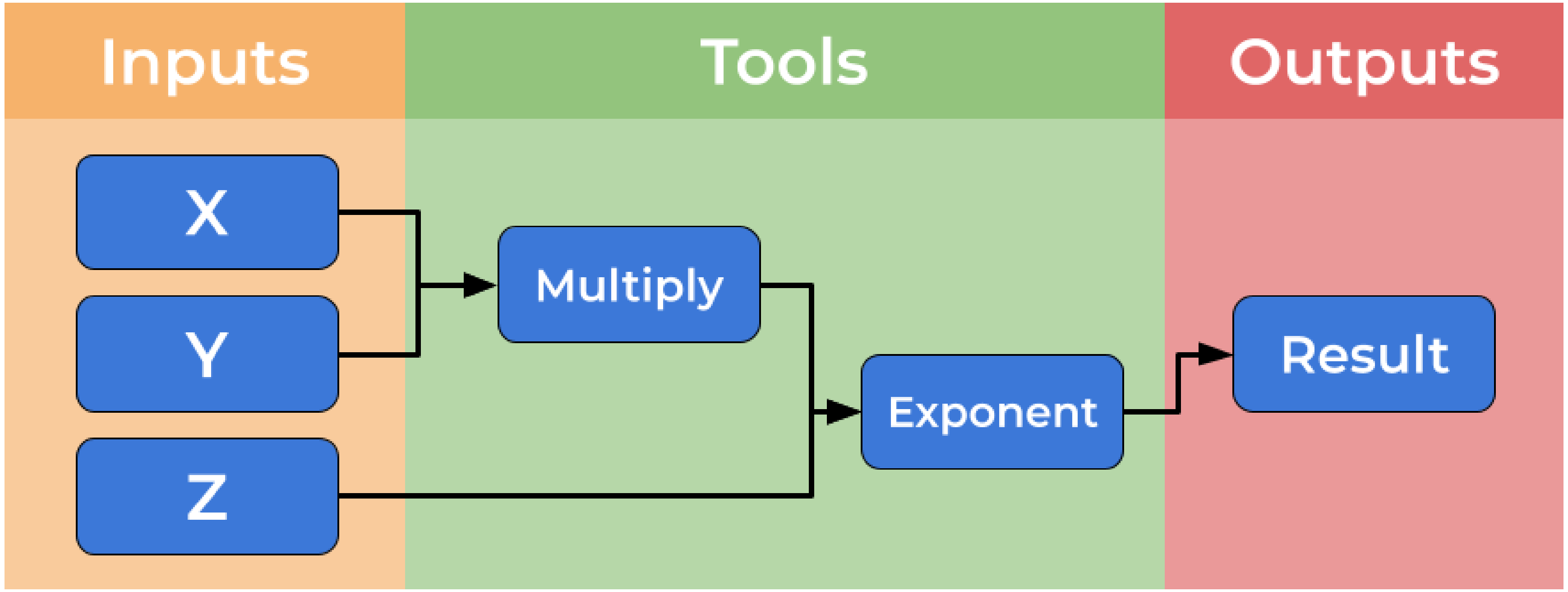


Introduction



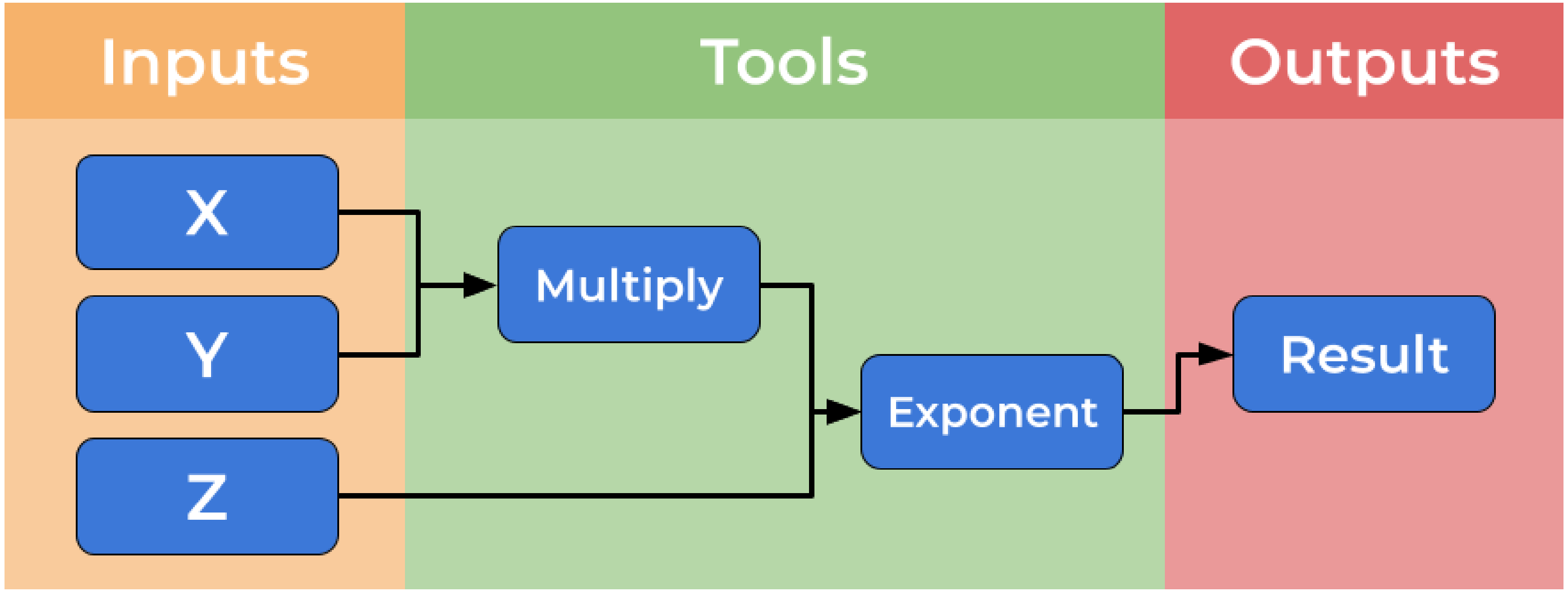
What is a workflow?

Computational workflows explicitly create a divide between a user’s dataflow and the computational details which underpin the chain of tools, placing these elements in separate files



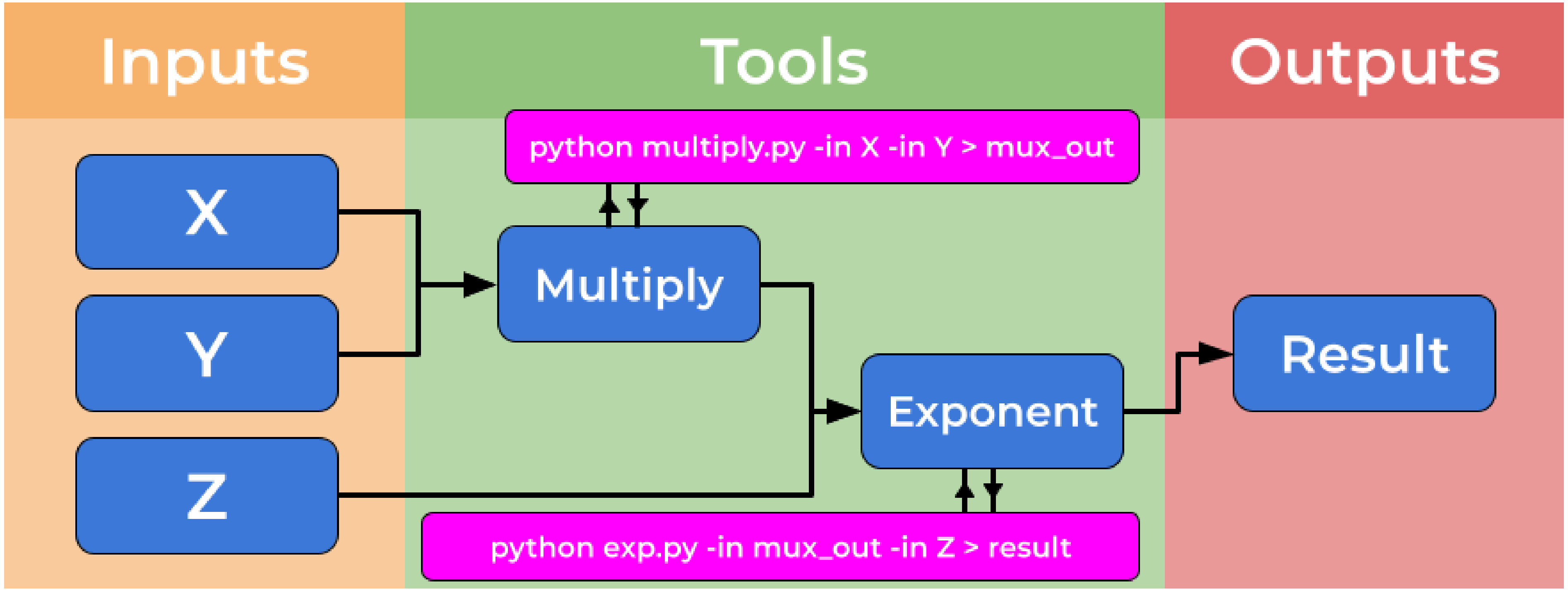
What is a workflow?

Control flow: Dictate the sequence in which steps are executed (Analogous to traffic lights & intersections)
Data flow: *what* moves between steps (eg: files, data) (Analogous to cars)

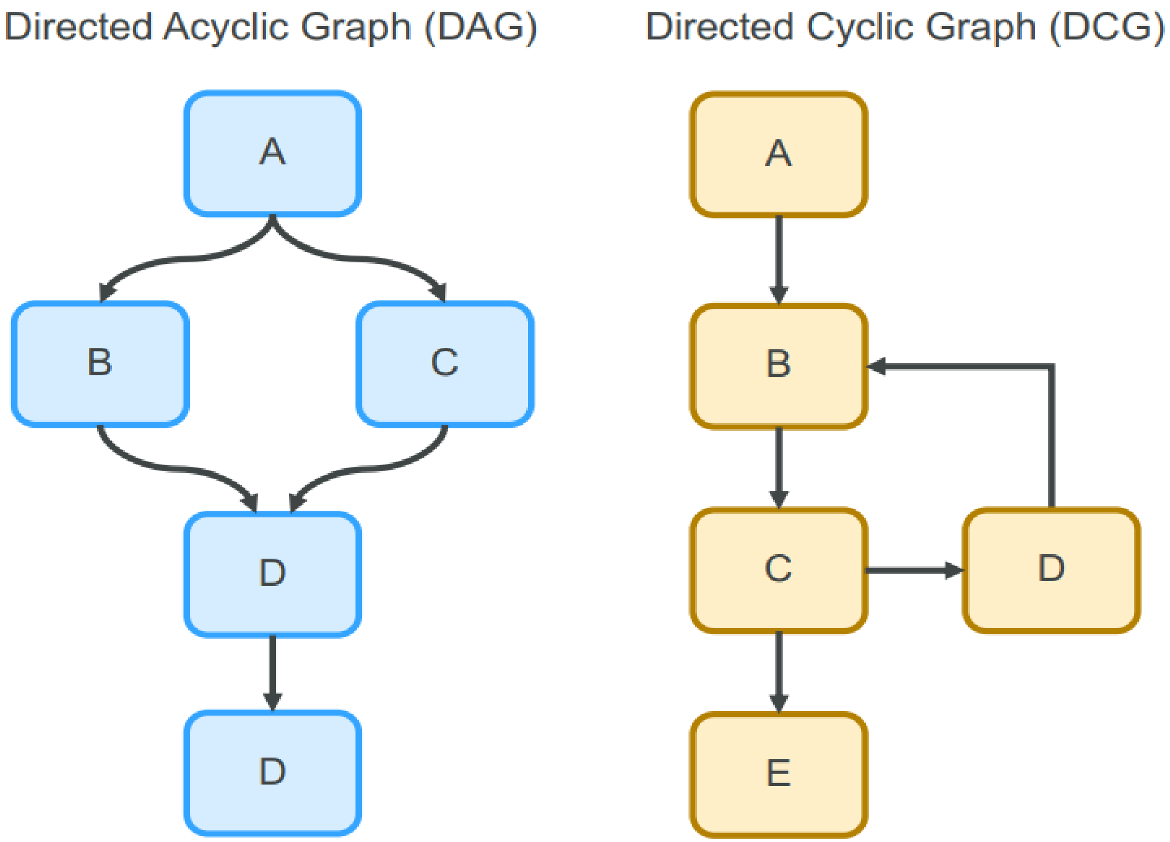


What is a workflow?

Abstraction: components can be thought of as black boxes



Complex workflows



DAGs are the basis for most workflow engines:

- Apache Airflow
- Luigi
- DAGster
- …

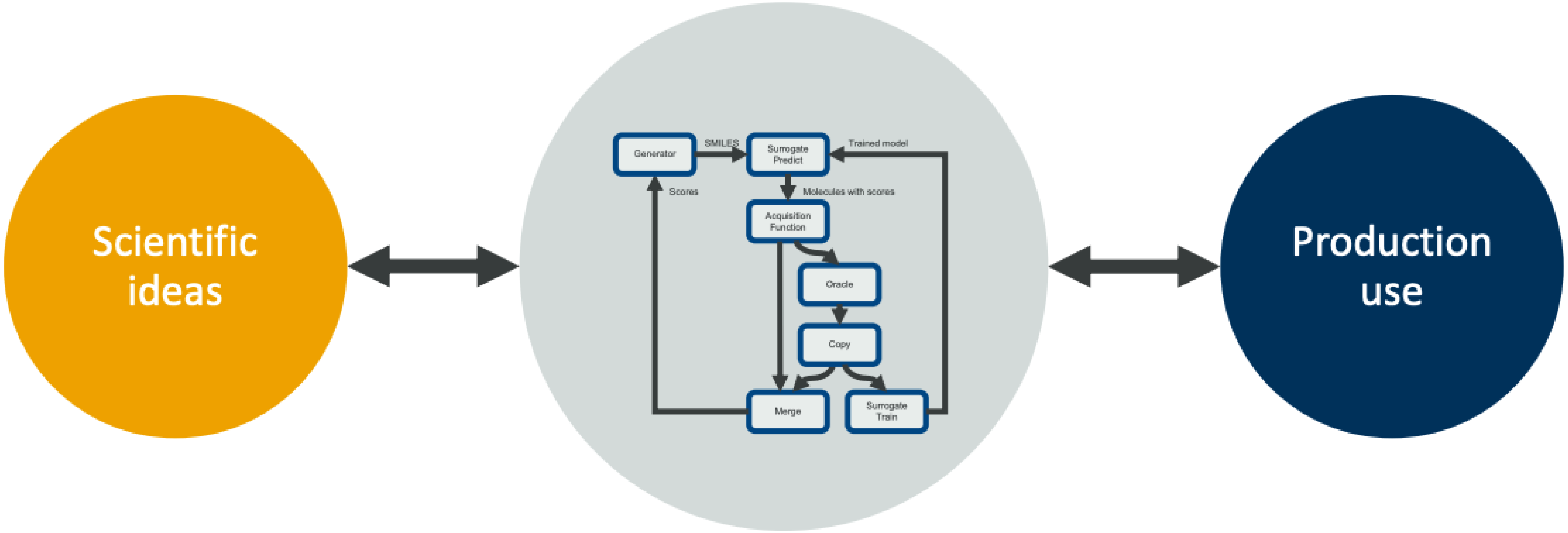
DCGs allow iteration, control flow, but have limited support:

- Akka (Scala)
- NoFlo (JavaScript)

What are workflow managers?

Workflow managers are a tool for abstraction!

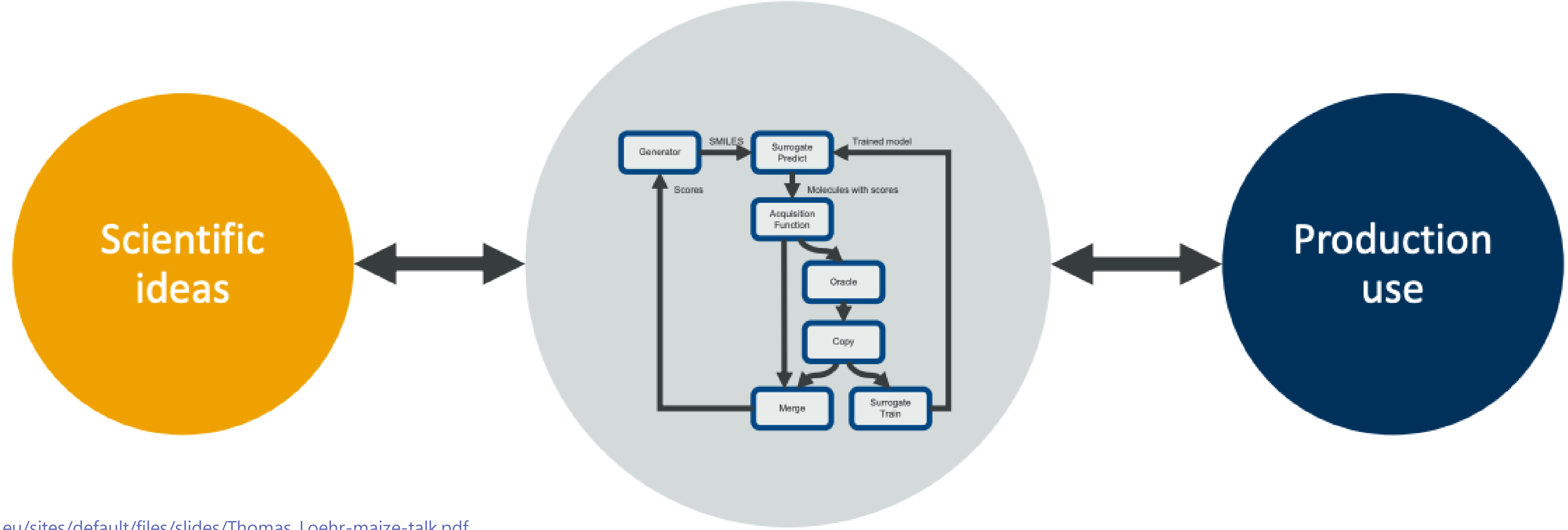
you write “what” to run; the system handles “how/where/when”



https://ai-dd.eu/sites/default/files/slides/Thomas_Loehr-maize-talk.pdf

What are workflow managers?

relax → phonons → DOS over ~200 structures, with automatic plane-wave



https://ai-dd.eu/sites/default/files/slides/Thomas_Loehr-maize-talk.pdf

Commonly used workflow managers

Nextflow

A DSL for data-driven computational pipelines

</> Groovy
Version 24.04.2
04 Jun 2024
WCI metadata

AiiDA

A workflow manager for computational science with a strong focus on provenance, performance and extensibility.

</> Python
AiiDA v2.5.0
01 Jun 2024
WCI metadata

pyiron

A workflow manager for scientific computing on high performance computing infrastructures

</> python
pyiron_base 0.9.0
04 Jun 2024
WCI metadata

Parsl

Productive parallel programming in Python

</> Python
1.2.0
03 Jun 2024
WCI metadata

100+ workflow managers available…

Why not use python or bash scripts?

- **Reproducibility** – not just for others
- **Configuration** – no searching through shell history for used parameters
- **Modularization** – easily exchange components to make experiments easier
- **Automation** – easier to integrate into routine systems
- **Abstraction** – components can be thought of as black boxes
- **Performance** – some workflows allow parallelization

Part 2

Introduction to AiiDA: automated interactive infrastructure and database for computational science

AiiDA

Automated Interactive Infrastructure and Database for Computational Science

ADES

Automation	Data	Environment	Sharing
Remote management Coupling to data High-throughput	Storage Database Provenance	High-level workspace Scientific workflows Data analytics	Social ecosystem Repository pipelines Standardization

Language: implemented and API in Python

License: MIT open source **http://www.aiida.net/**

Source: https://github.com/aiidateam/aiida-core

MIT LICENSED

open source initiative

AiiDA plugins

QUANTUMESPRESSO, CP2K, VASP, Fleur, siesta, CASTEP, CRYSTAL17, NWCHEM, JuKKR, Yambo, OLGGLMU, WANNIER90, LAMMPS

raspa2, zeo++, ...

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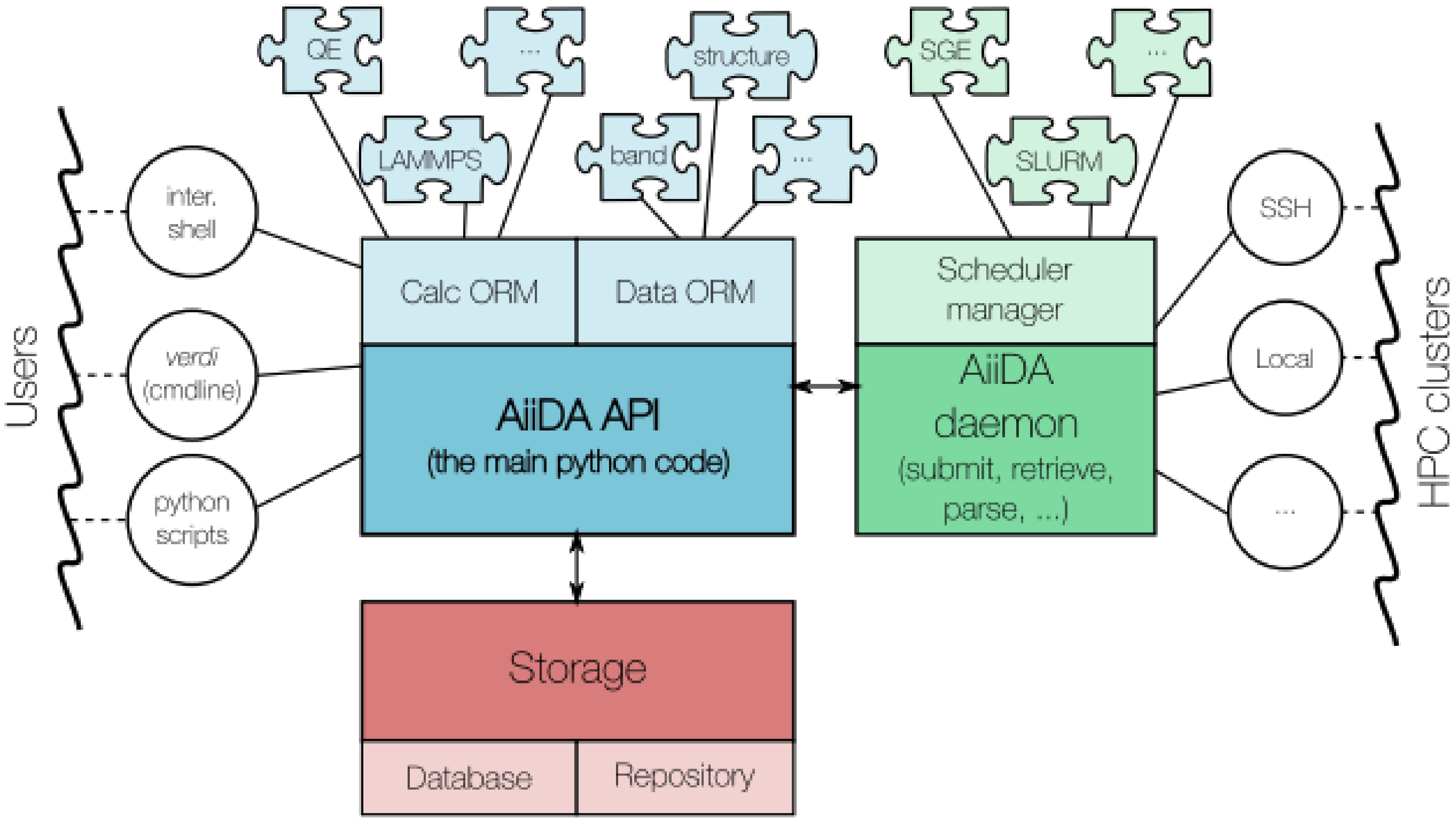
AiiDA schema



Object–Relational Mapping: a layer that maps **classes/objects** in code to **tables/rows** in a relational database.

Plugin interface: supports any computational code and data type via extensible plugins.

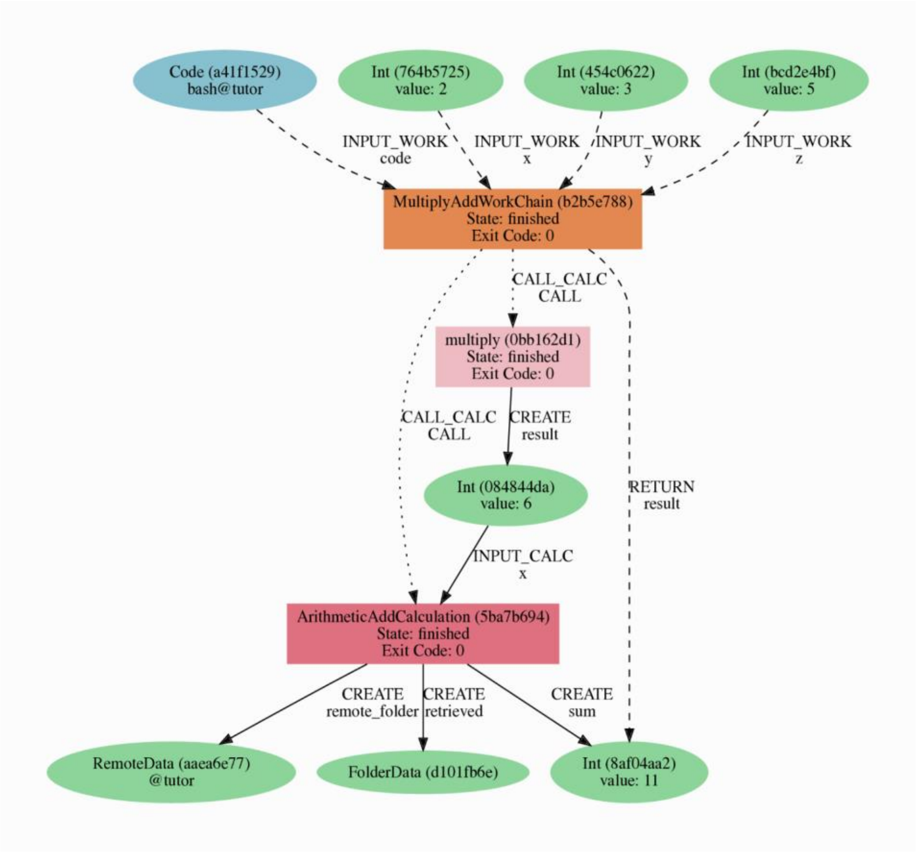
Daemon (background engine): automates job submission, monitoring, file retrieval, and parsing.

Schedulers: talks to SLURM/PBS/… through scheduler plugins; abstracts queue details.

Transports: connects to resources locally or via SSH (and other channels).

Remote interaction: submits jobs, checks states, and fetches results from clusters automatically.

Provenance



Provenance graph

- AiiDA stores **results, inputs, and every execution step** of your calculations.
- All this information is captured as a **Directed Acyclic Graph (DAG)**.
- Provenance lets you **fully retrace how any data was produced → reproducibility**.

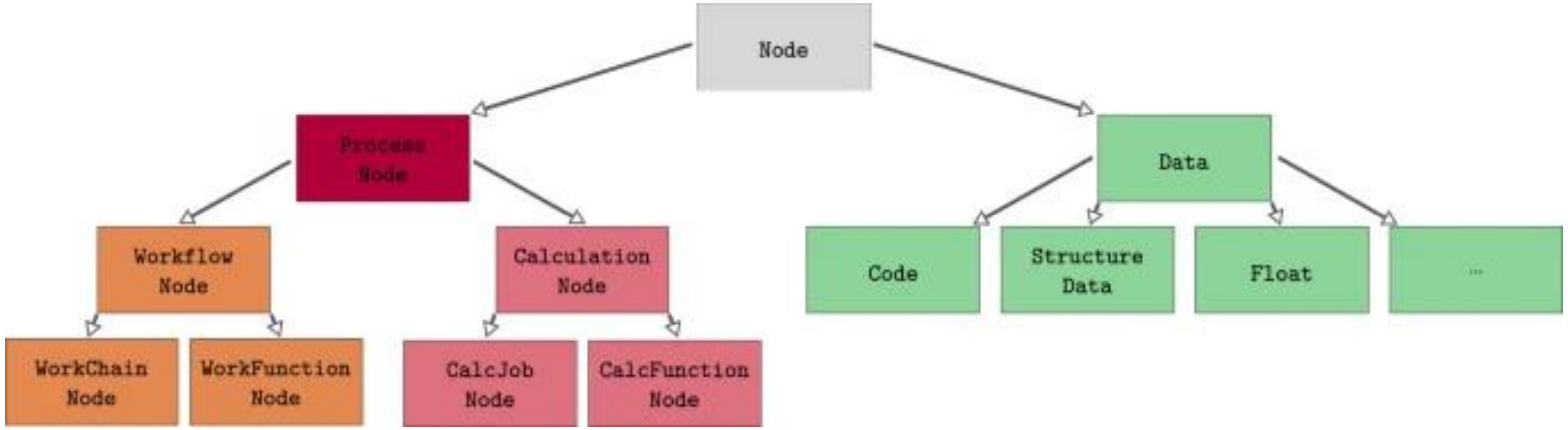
Basic concepts

Data Types

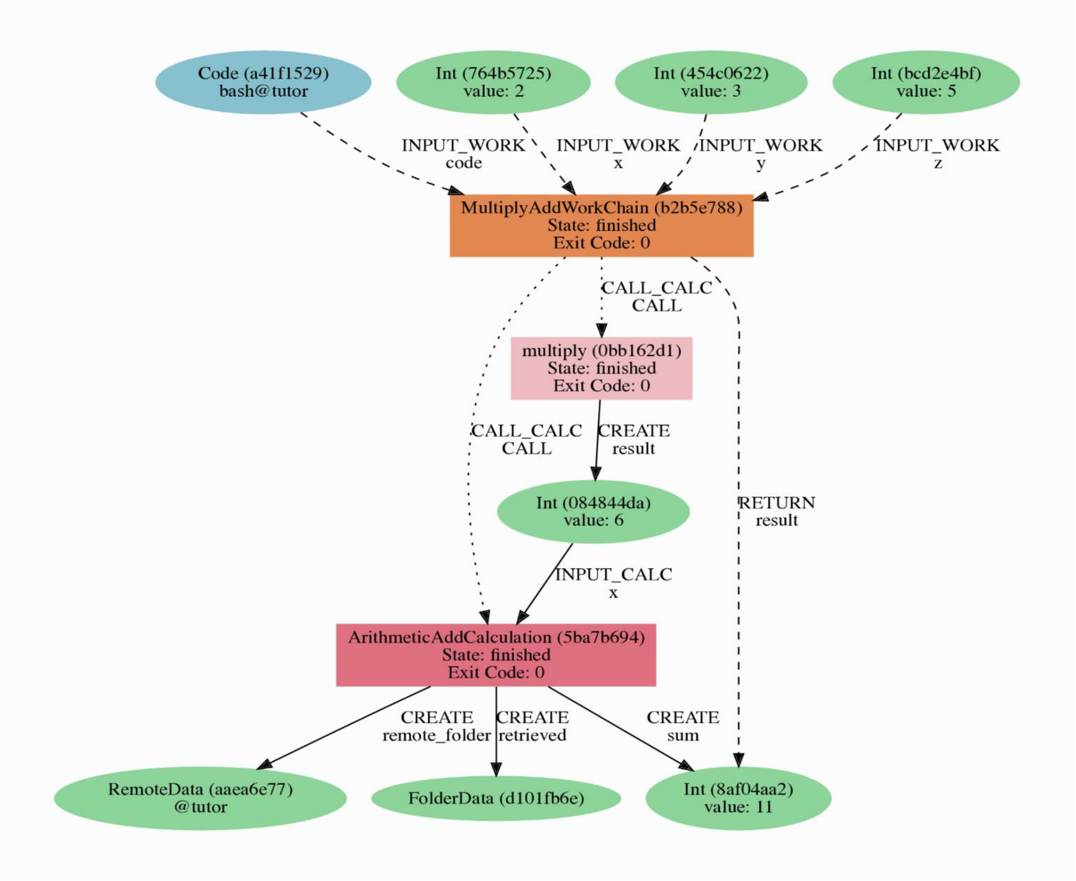
- Python classes hosting data, allow storing in the database and provenance.
- Simple wrappers of python types.

```
from aiida.orm import Float
vols = Float(7)
```

```
from aiida.orm import StructureData
struct = StructureData(ase=ase_struct)
```



Hands on



📘 AiiDA Tutorial

You can open the tutorial notebook here:

👉 AiiDA Tutorial (Google Colab)

📝 AiiDA Exercise

After completing the tutorial, open the exercise notebook here:

👉 AiiDA Exercise (Google Colab)

🚀 How to Use

1. Click the links above to open the notebooks in **Google Colab**.
2. Make sure you are signed into your Google account.
3. Select "**Copy to Drive**" to create your own editable version.
4. Follow the instructions inside each notebook.

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